



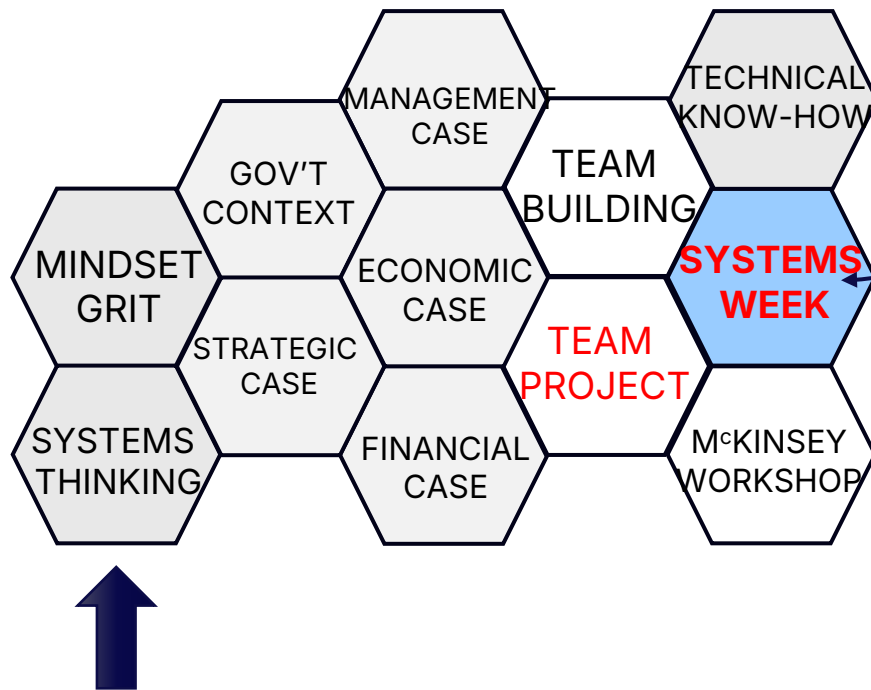
Waipapa
Taumata Rau
**University
of Auckland**

Systems thinking and wicked problems

Lecture 8

➤➤➤ August 2026

➤➤➤ Juliet Gerrard and Amanda di Ienno



Crazy sounding assessment
(until you hit the real world)



Outline

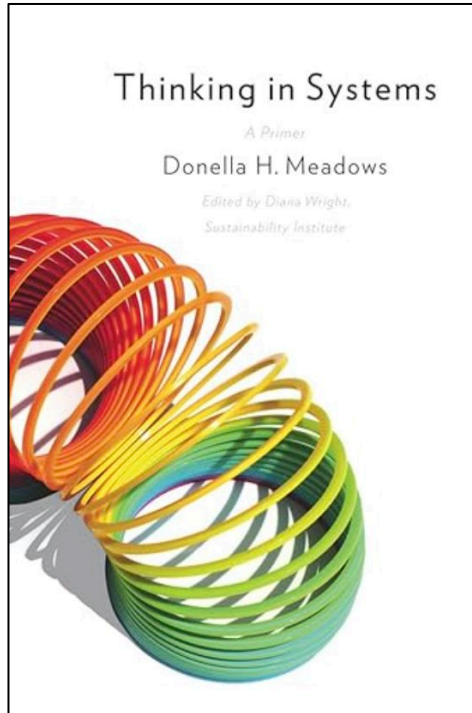
- Systems thinking - recap from lecture 2
- A brief history of systems thinking
- Wicked problems and our VUCA future
- Example: tackling obesity
- Systems thinking for wicked problems
- Reminders



Learning outcomes

- Be familiar with the history and foundations of contemporary systems thinking
- Characterise and contextualise systems thinking as both a discipline and a set of useful tools for addressing complex and wicked problems
- Describe the characteristics of wicked problems and why they cannot be solved, only minimised
- Identify complex systems and scope problems at scale
- **Identify and** apply tools for analysis of complex systems and tackling wicked problems

Reading: (at least the) first 2 chapters of *Thinking In Systems*

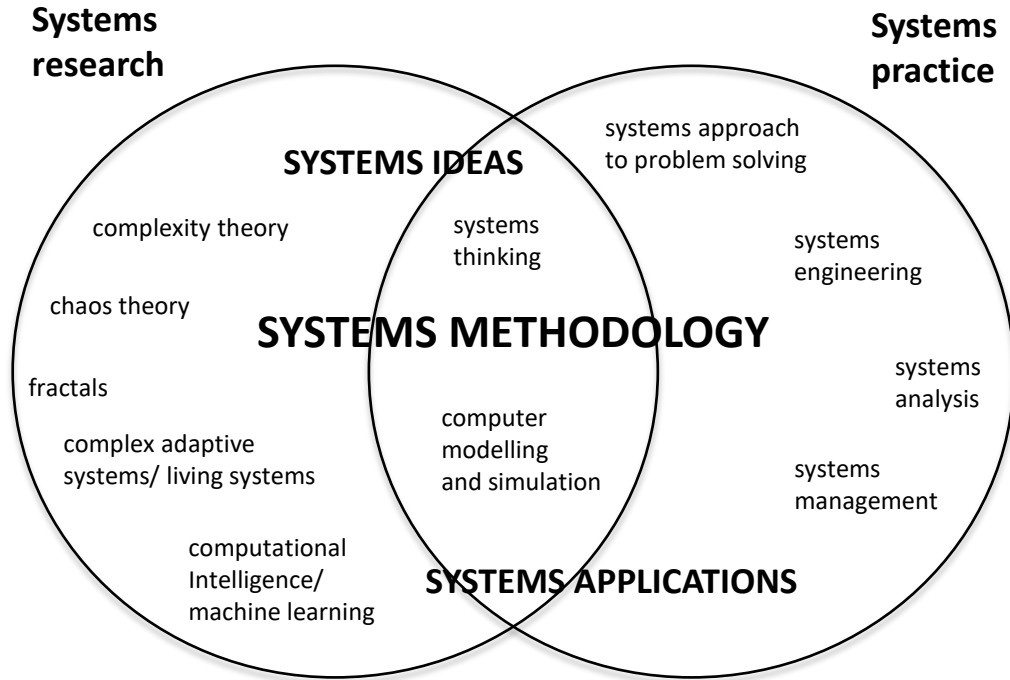


Reading goals:

- Hone your abilities to recognise systems and their parts
- Learn to look for interconnections
- Learn to ask "what if" questions about possible future behaviours
- Be creative and courageous
- Think about system design, and re-design

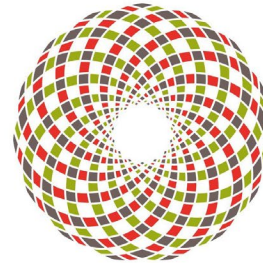
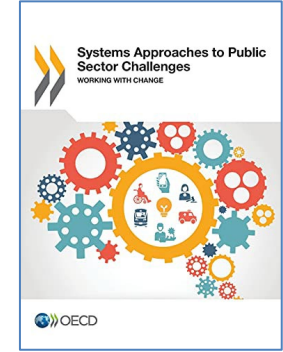
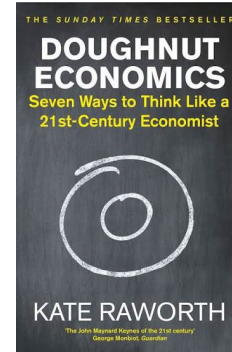
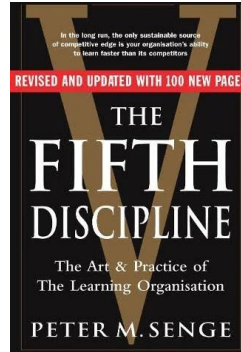
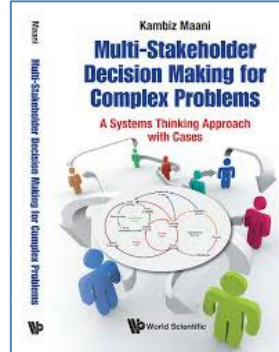
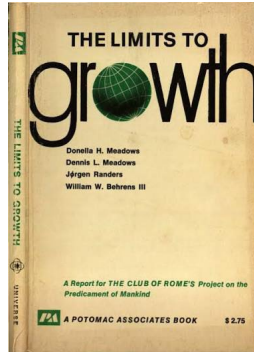


From lecture 2

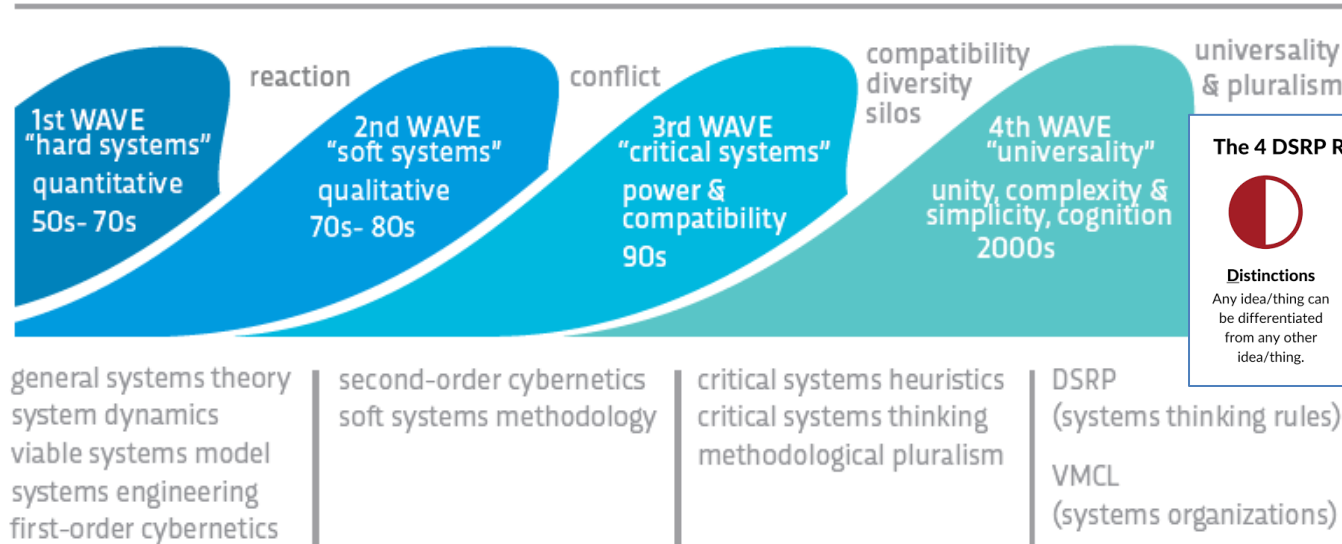


“the biggest impediment to progress comes, **not from the engineering side** of industrial problems, but from the management side ... social problems are much harder to understand and control than physical systems”

Portland State



Te Pūnaha Matatini
Complexity is at our heart



The 4 DSRP Rules of Systems Thinking

READINGGRAPHICS
Ideas Come Alive



Distinctions

Any idea/thing can be differentiated from any other idea/thing.



Systems

Every idea/thing can be split into sub-parts, or be part of a larger whole.



Relationships

Any idea/thing can relate to other ideas/things.

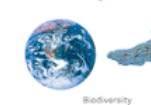


Perspectives

Any idea/thing can be seen from the perspective of another idea/thing.

VMCL: Build an Adaptive Organization

VISION: concise, measurable future state or goal?



Biodiversity

MISSION: simple rules, known by all, that in repetition lead to vision?



Go forth and multiply

CULTURE & CAPACITY: shared mental models that support vision, mission and management of mission critical systems?



LEARNING: constant adaptation based on feedback from the real world?



We humans are smart enough to have created complex systems and amazing productivity; surely we are also smart enough to make sure that everyone shares our bounty, and surely we are smart enough to sustainably steward the natural world upon which we all depend.

Donella Meadows

Our VUCA future: volatile, uncertain, complex, uncertain, volatile

- Technology
- Globalisation
- Demographics
- Climate change
- Geopolitical conflict



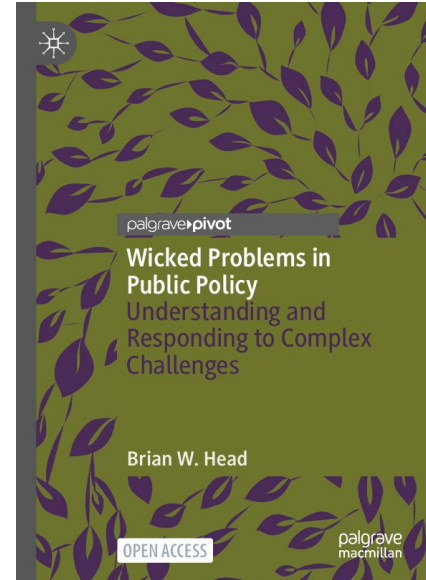
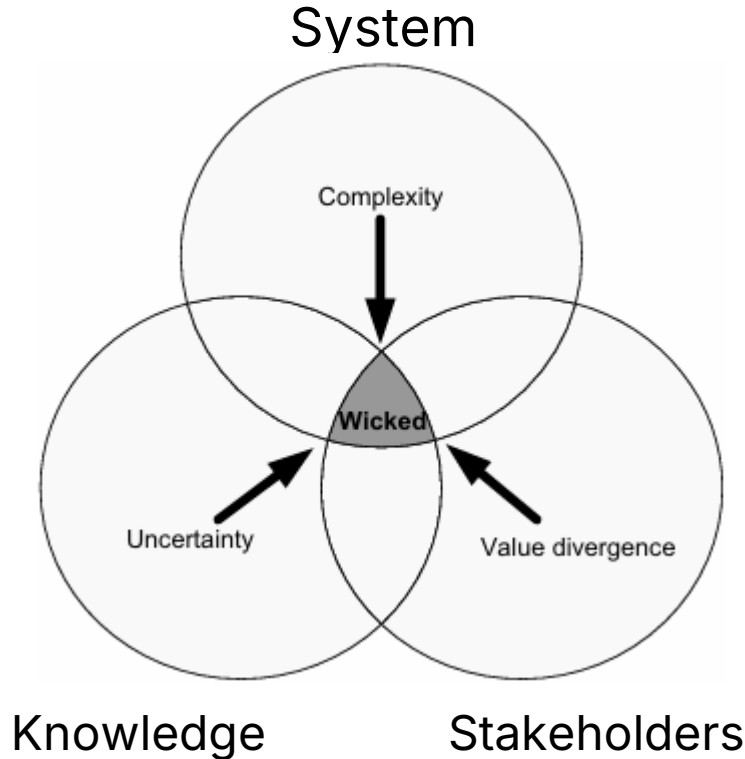
Wicked problems: unsolvable, undefinable, unverifiable, unique, uncertain

The term wicked problem is used in relation to issues that resist a complete or simple solution.

It is likely that wicked problems can be ameliorated but not solved, and that amelioration measures may create new problems.

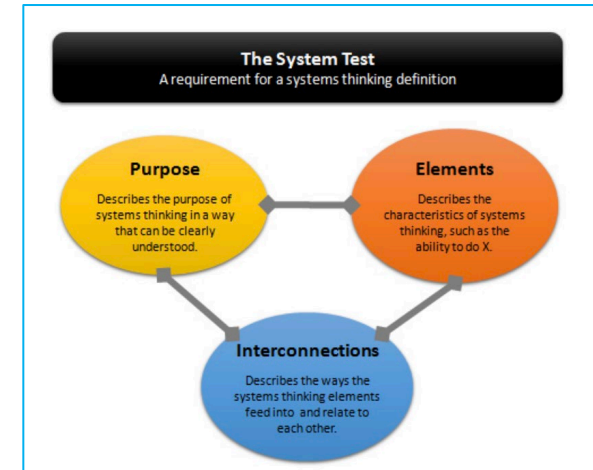
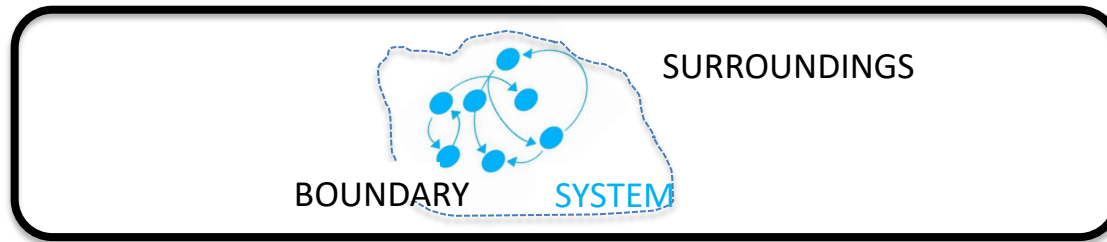
A wicked problem is distinct from a complex problem, which is hard, but may have an optimal solution, or a long problem, defined by its timeframe, although many wicked problems are often both complex and long.

Gerrard, Front Row Seat, 2026



example system definition 2

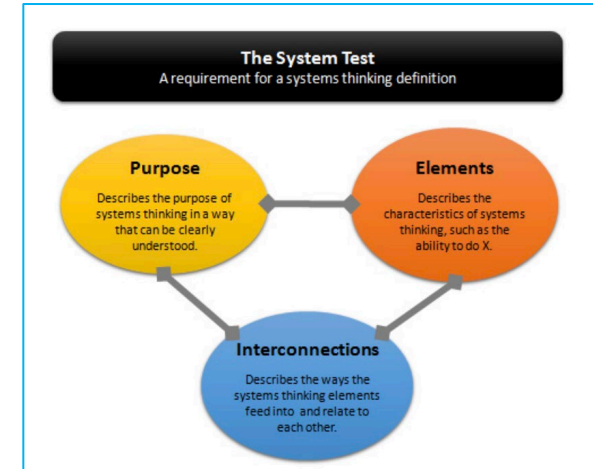
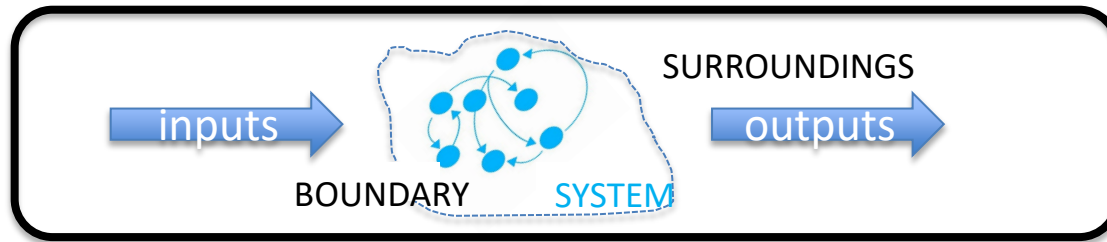
- A set of connected parts that together form a cohesive whole



- A system will have interconnected related elements, inputs, outputs, and a purpose

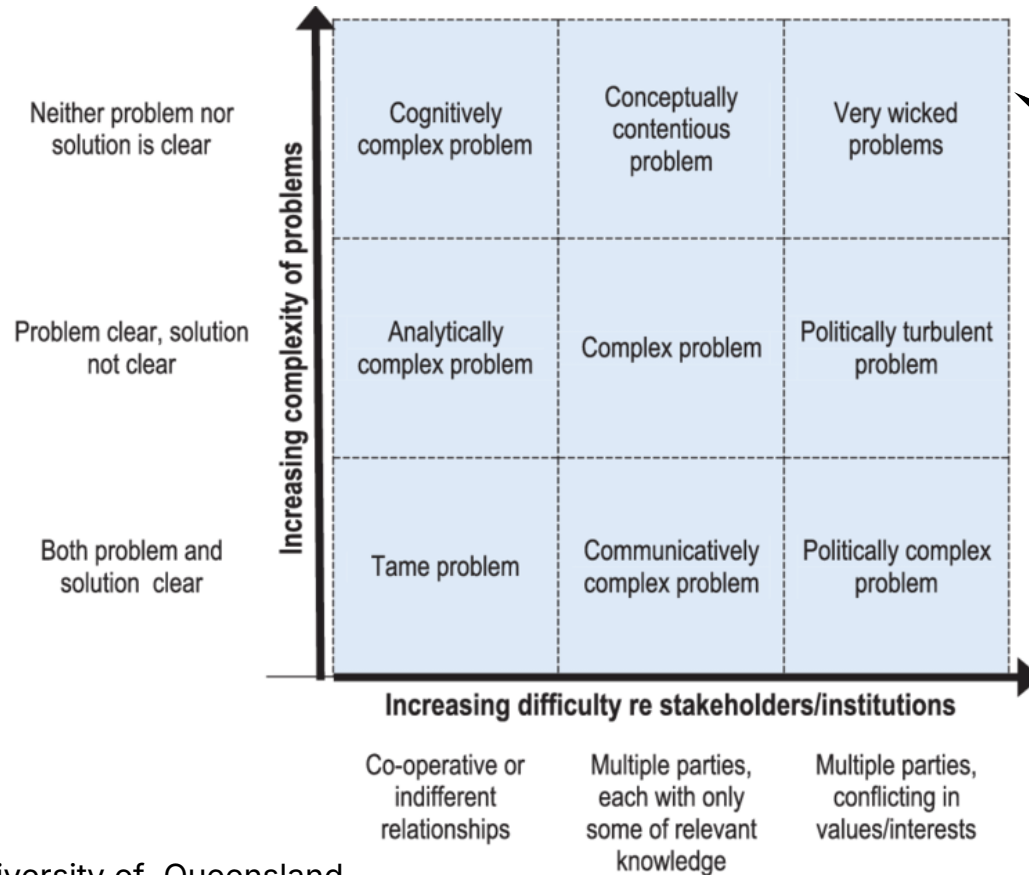
example system definition 2

- A set of connected parts that together form a cohesive whole

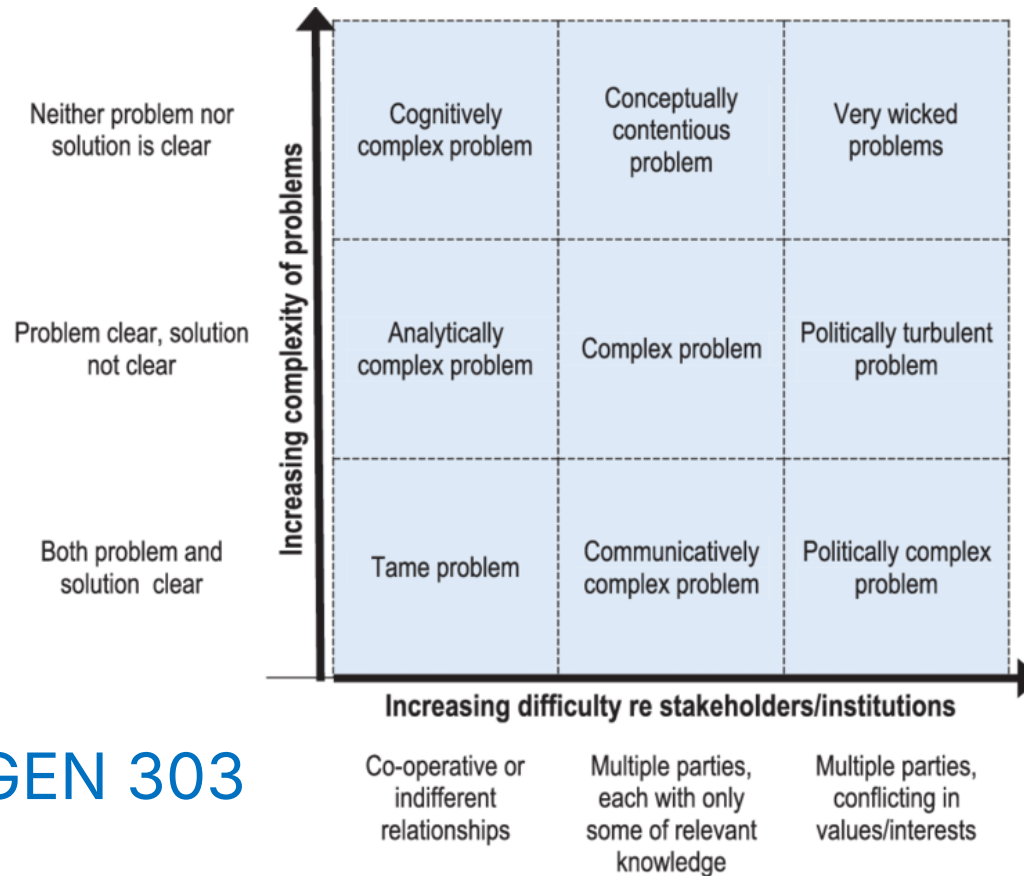


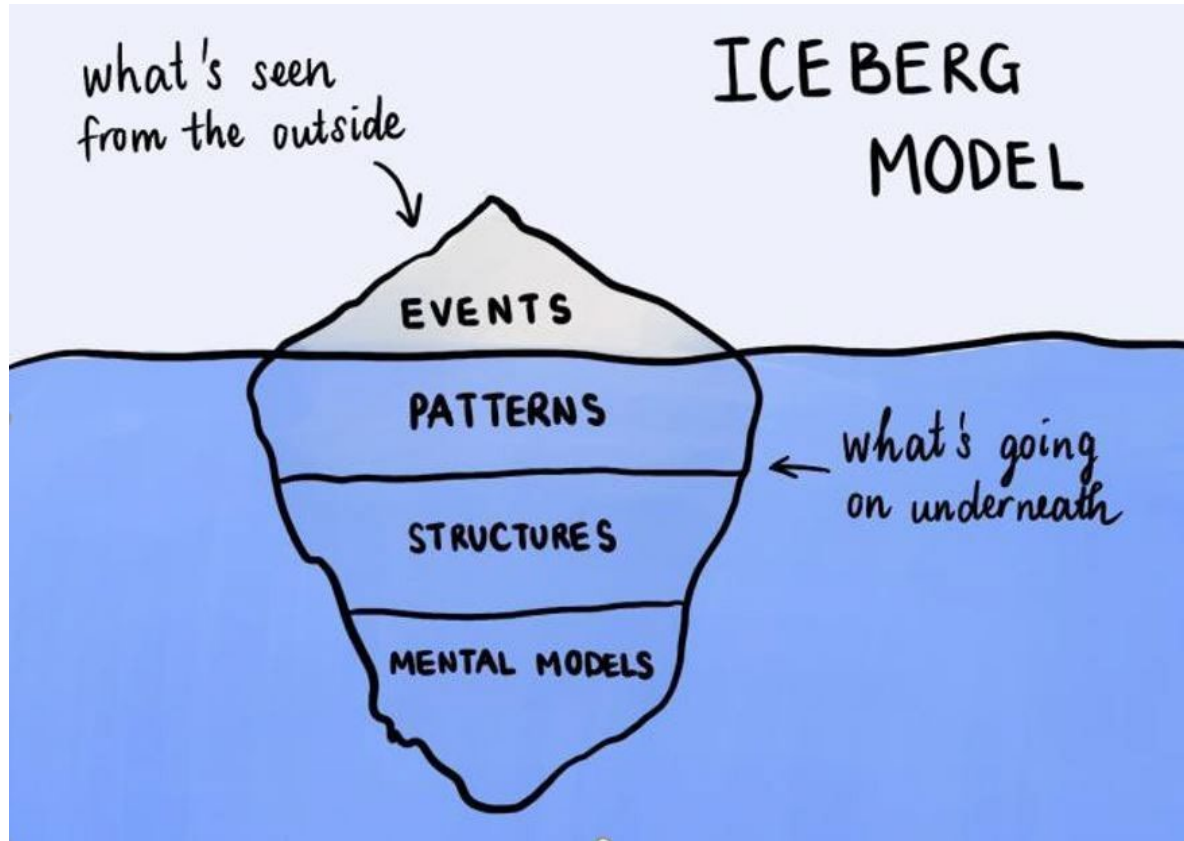
- A system will have interconnected related elements, inputs, outputs, and a purpose

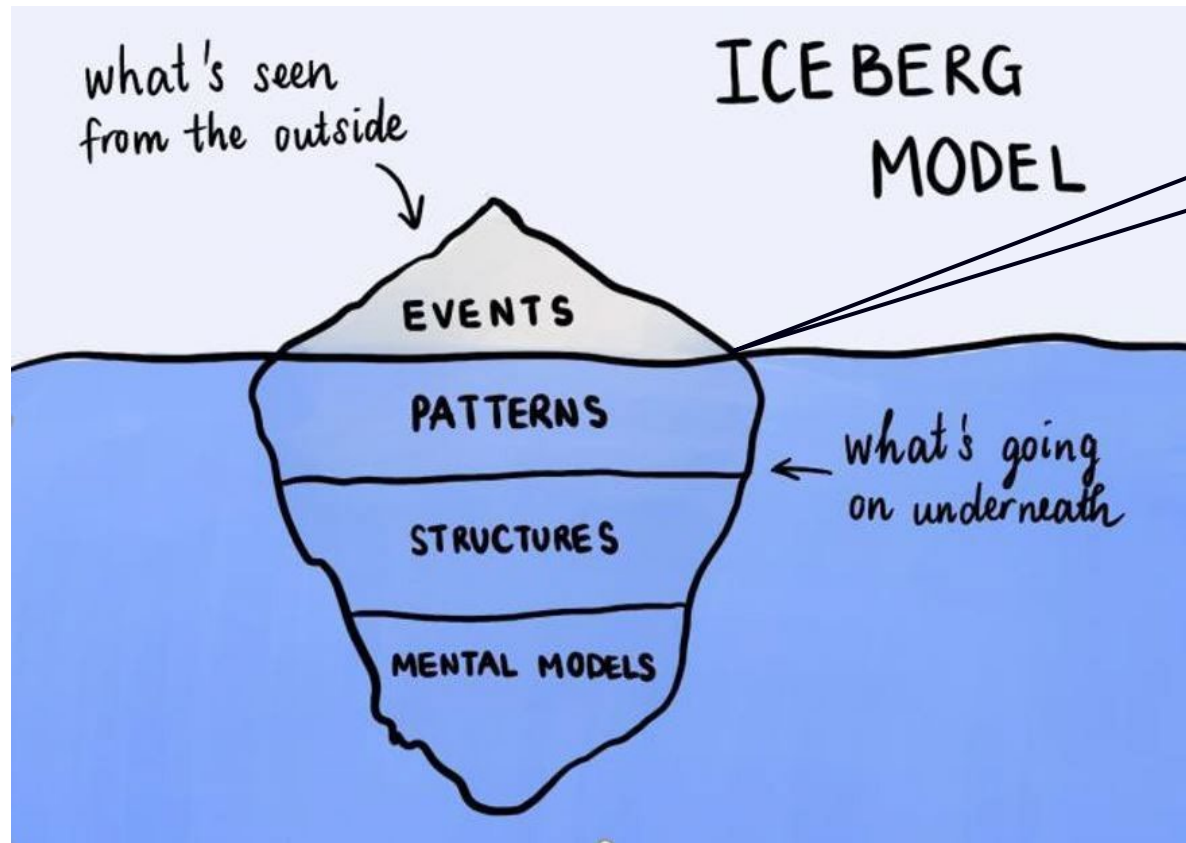
wicked vs complex



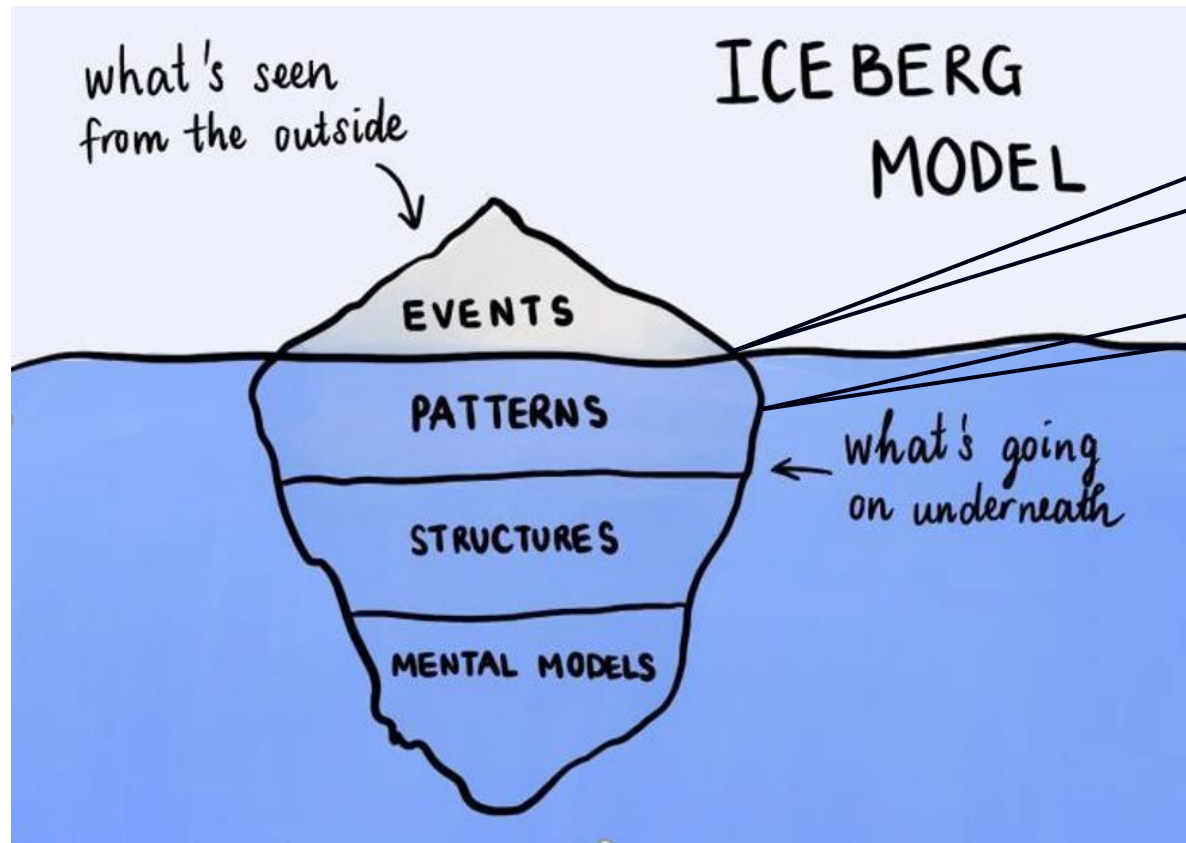
Complex issues involving many stakeholders with conflicting requirements





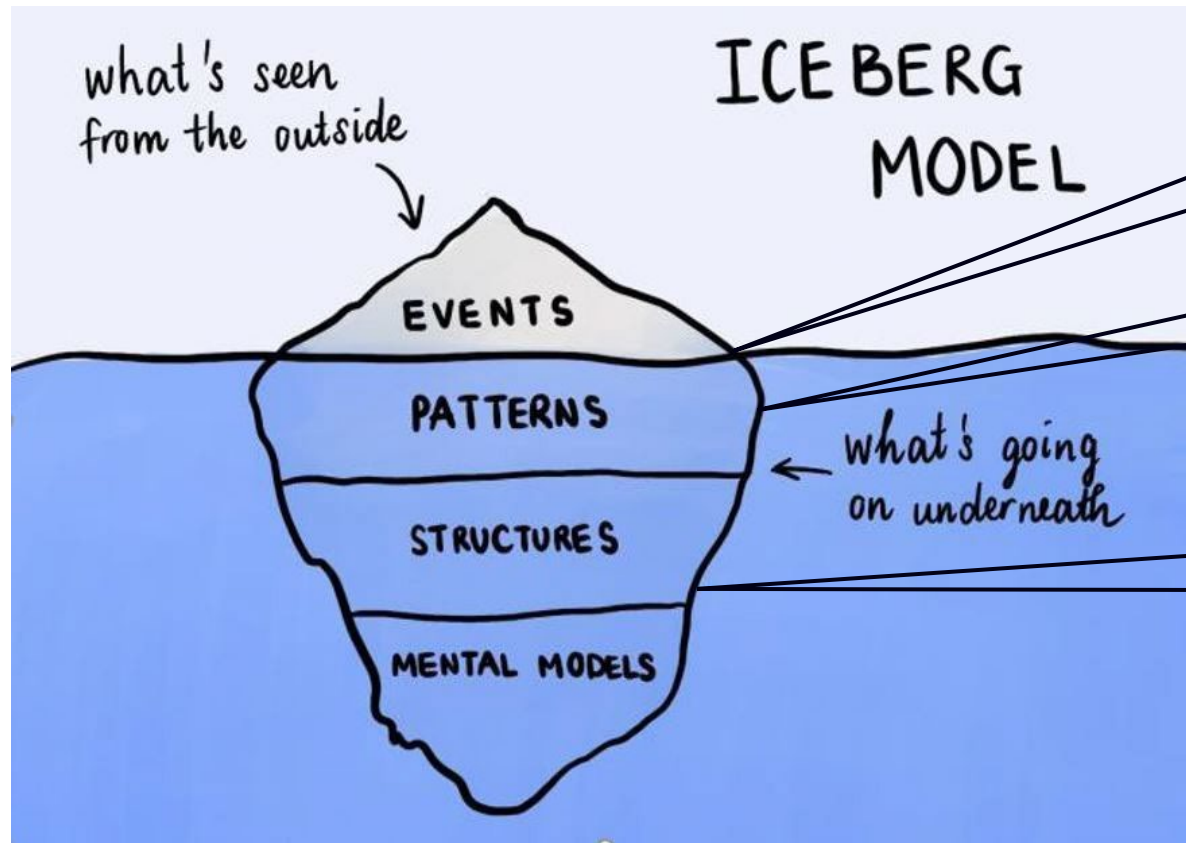


the level at which we typically perceive the world—for instance, waking up one morning to find we have caught a cold



the level at which we typically perceive the world—for instance, waking up one morning to find we have caught a cold

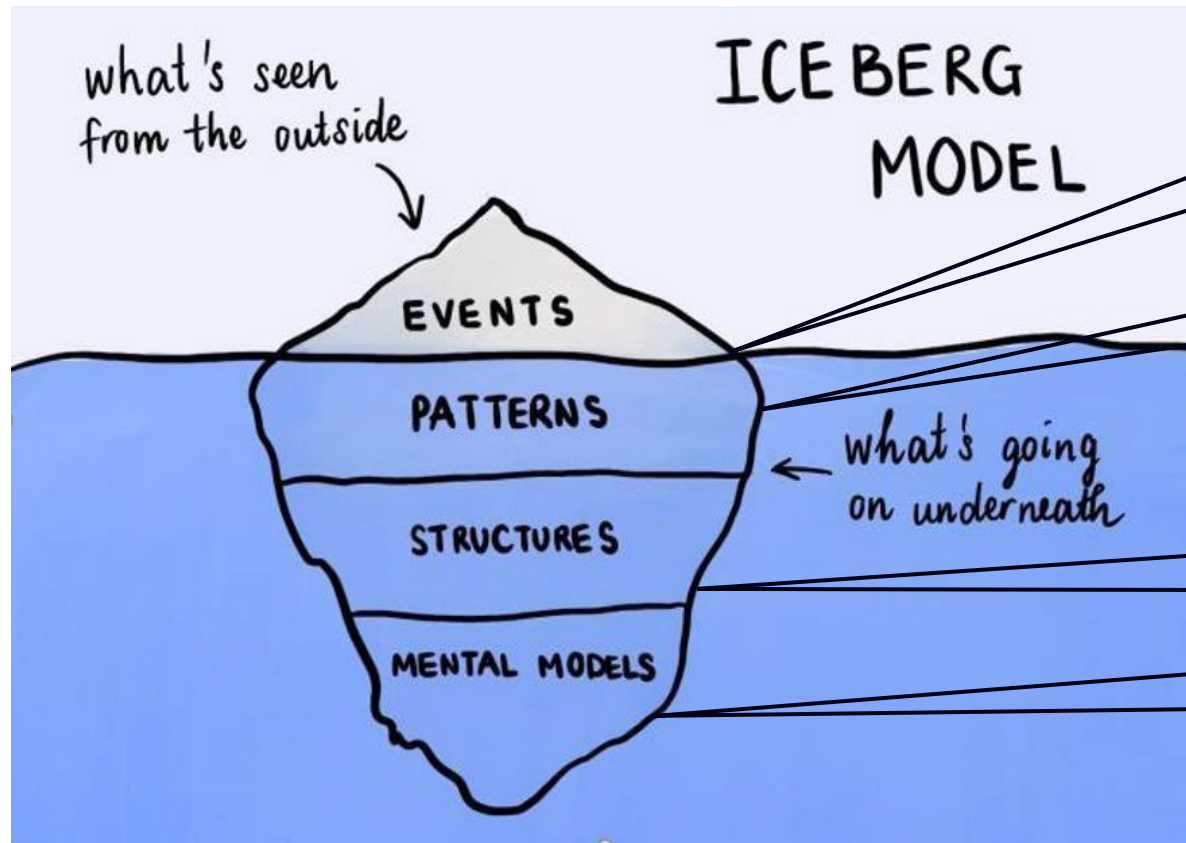
below the event level, we often notice patterns; similar events have been taking place over time



the level at which we typically perceive the world—for instance, waking up one morning to find we have caught a cold

below the event level, we often notice patterns; similar events have been taking place over time

below the pattern level lies the structure level, when we ask, "What is causing the pattern we are observing?"



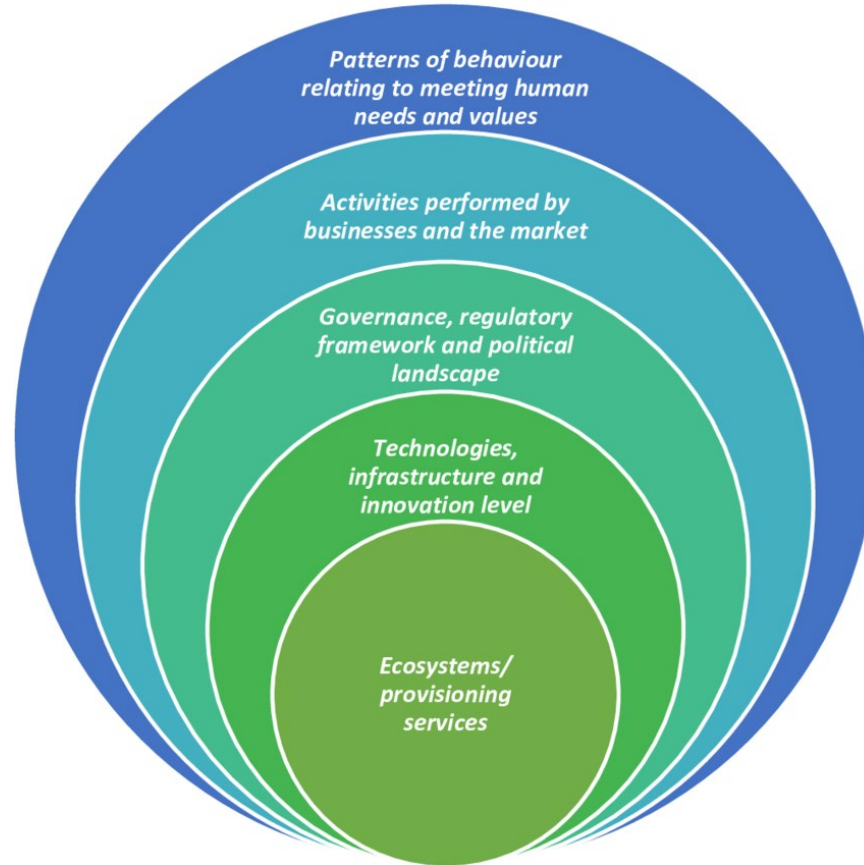
the level at which we typically perceive the world—for instance, waking up one morning to find we have caught a cold

below the event level, we often notice patterns; similar events have been taking place over time

below the pattern level lies the structure level, when we ask, "What is causing the pattern we are observing?"

mental models are the attitudes, beliefs, morals, expectations, and values that allow structures to continue functioning as they are

Levels of thinking



Eleni Iacovidou

Obesity



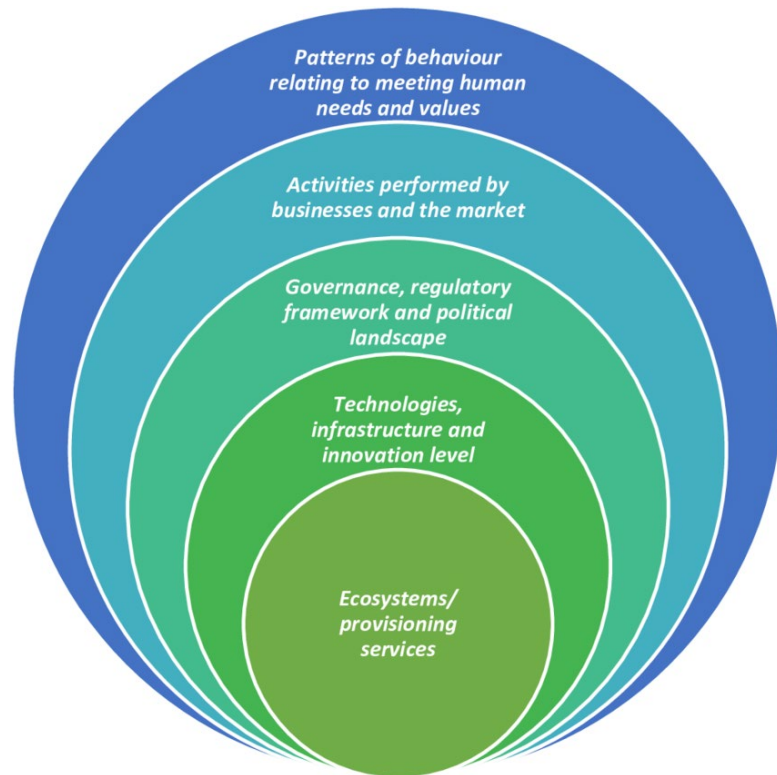
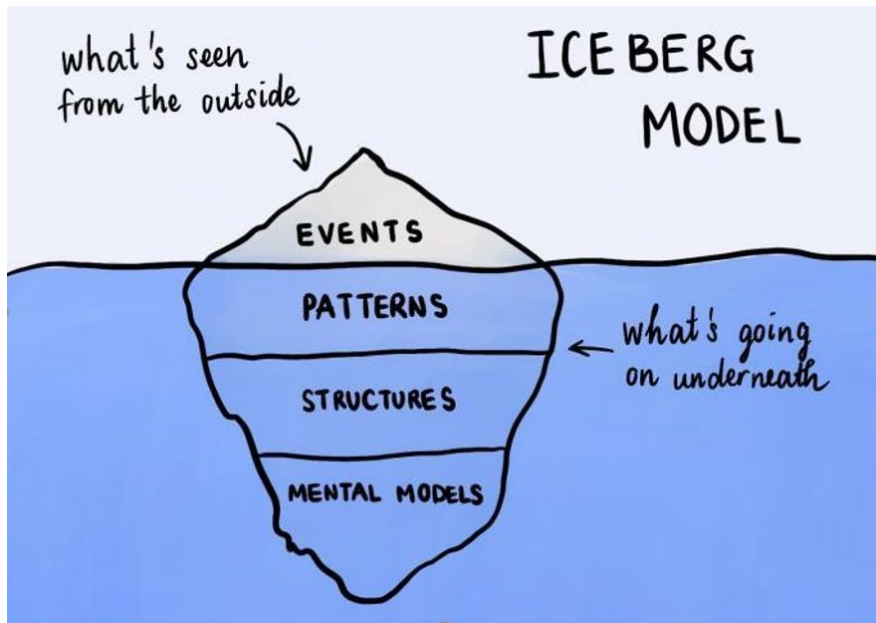
New Zealand adults who are classed as obese based on their body mass index

By gender, year ended June 2012-2025, % of adults

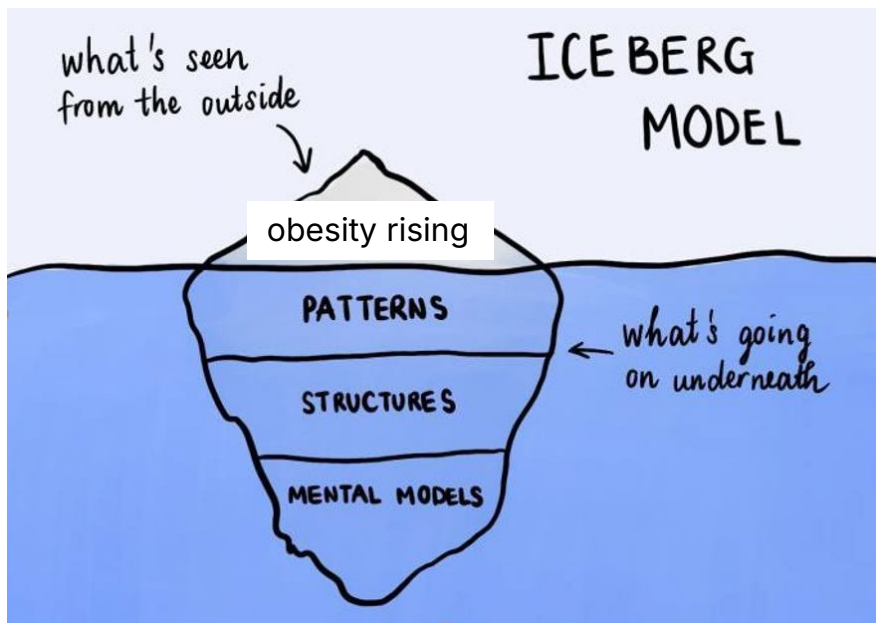
Provider: Ministry of Health



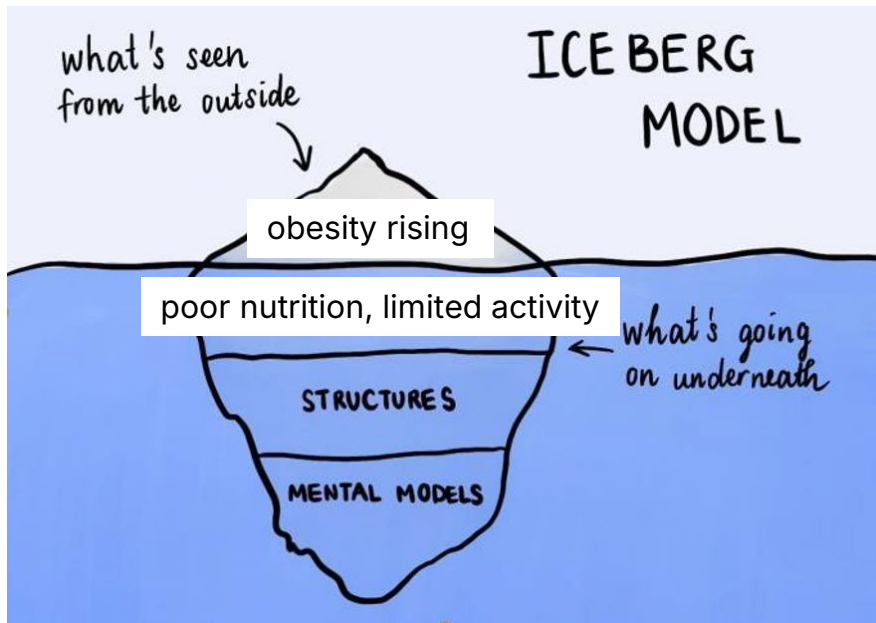
Obesity



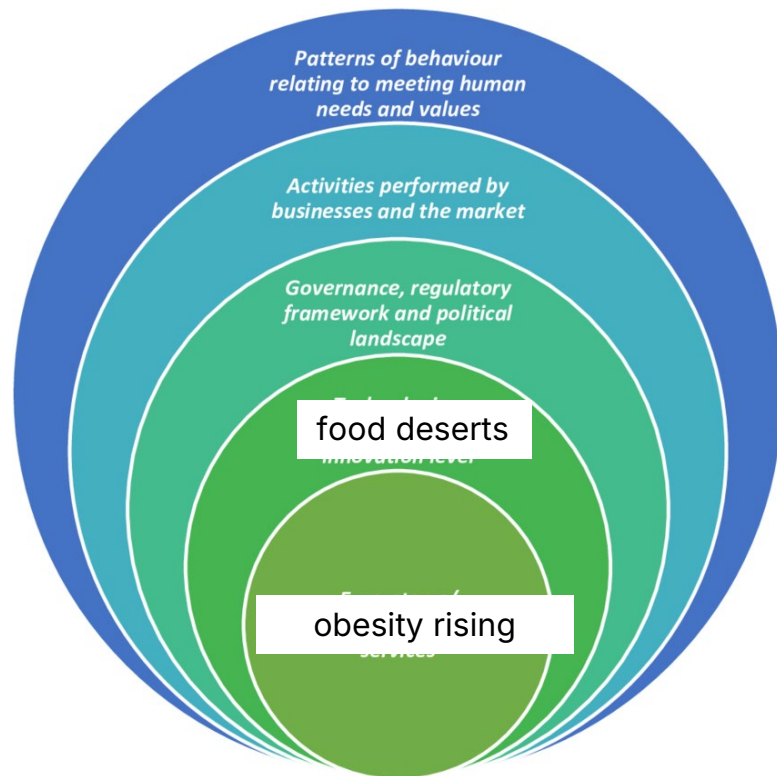
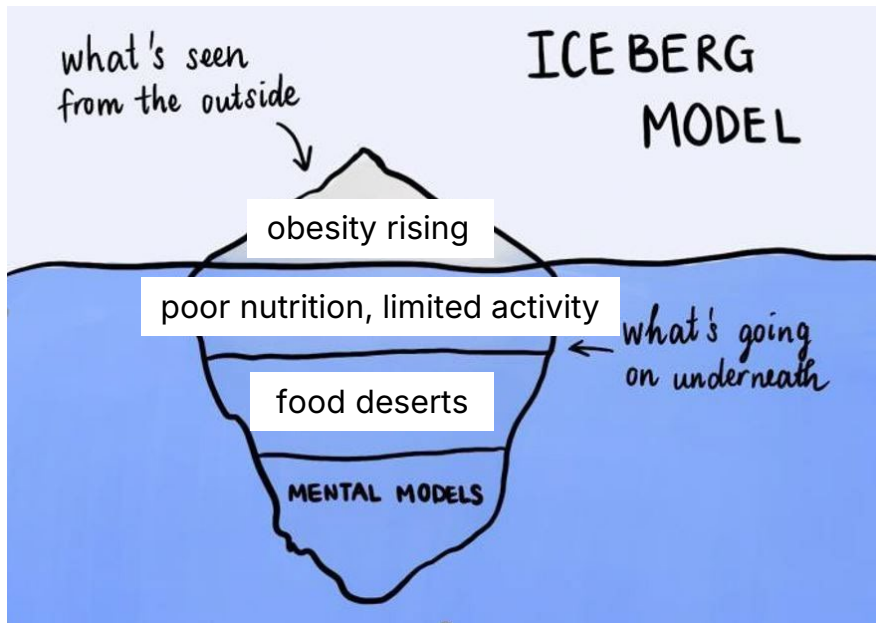
Obesity



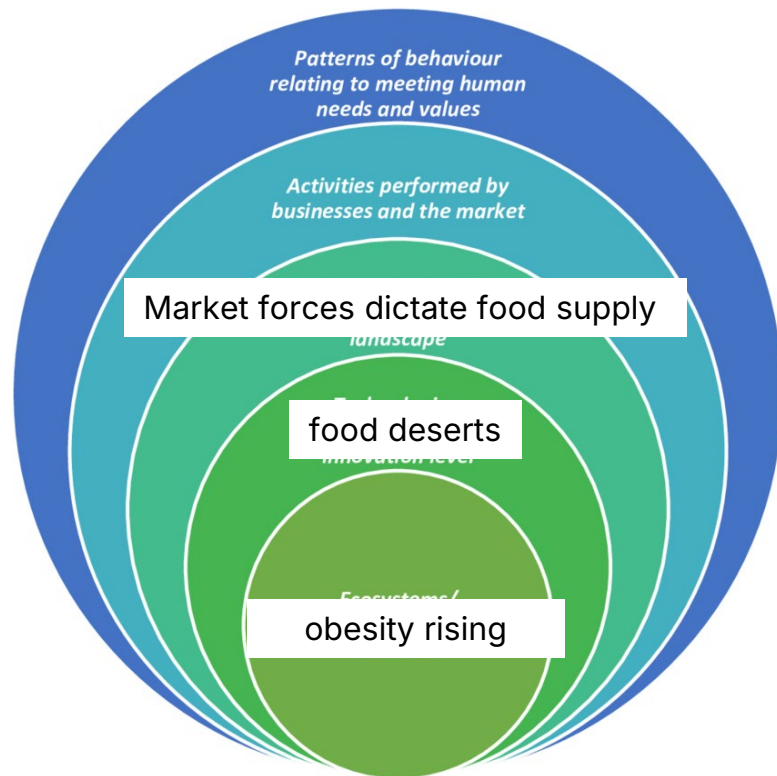
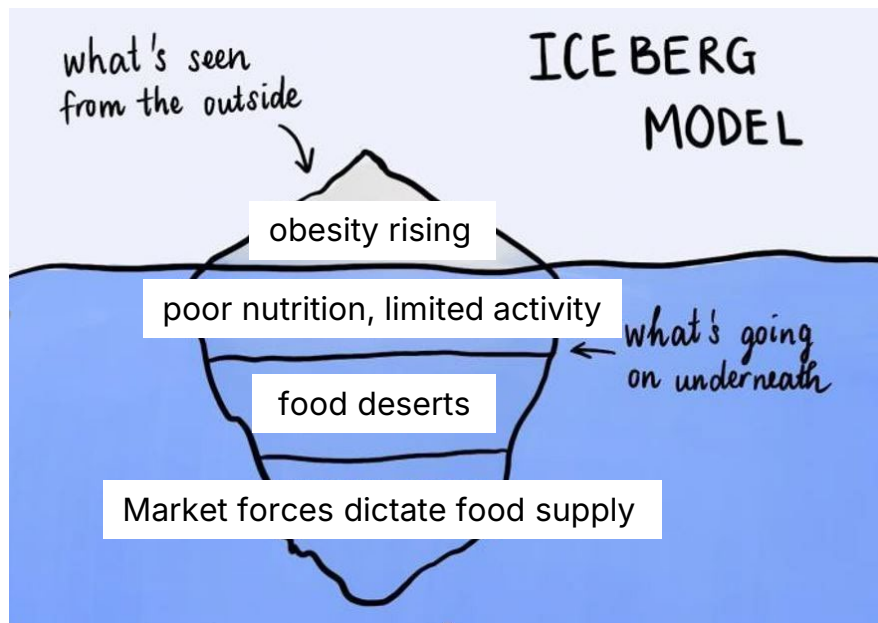
Obesity



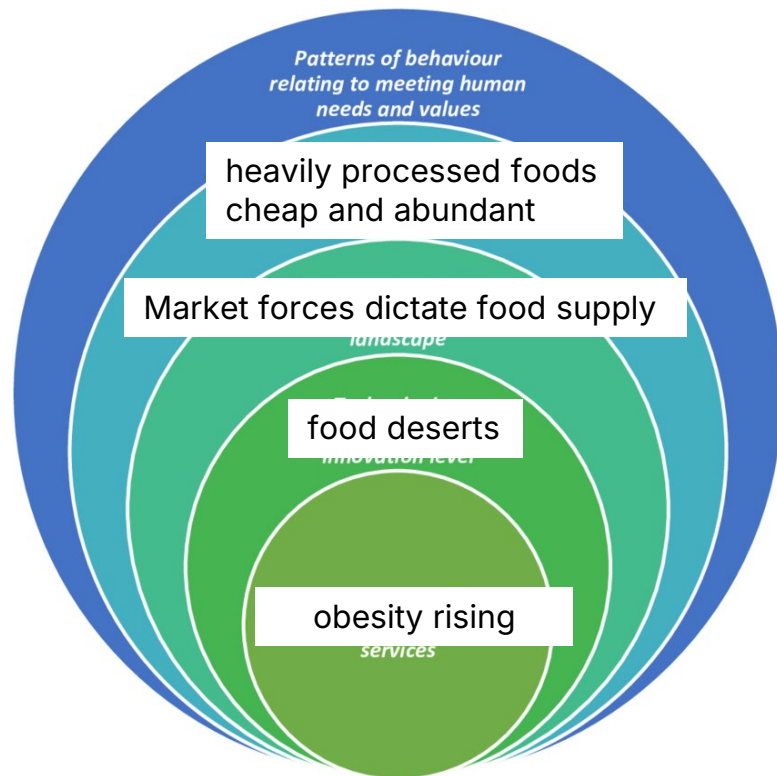
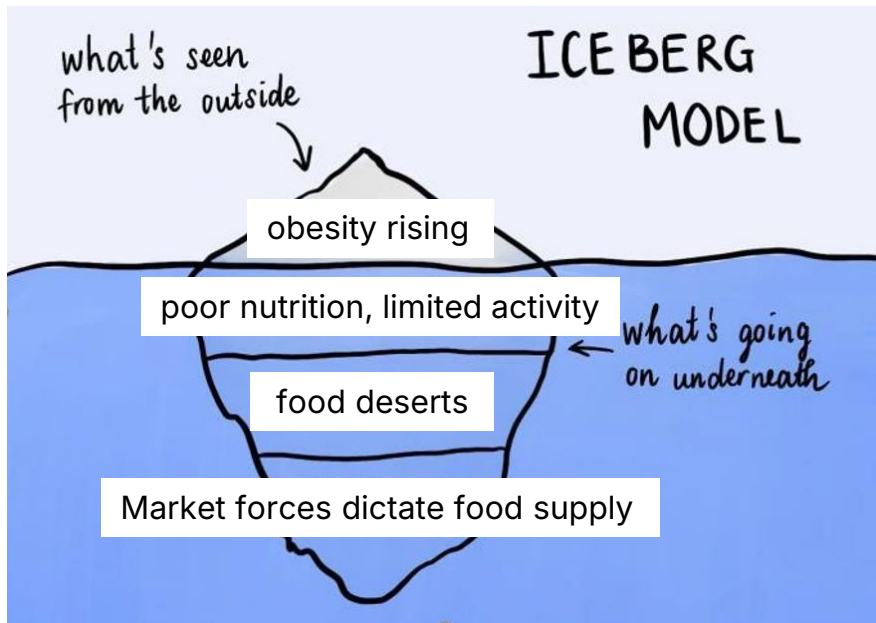
Obesity



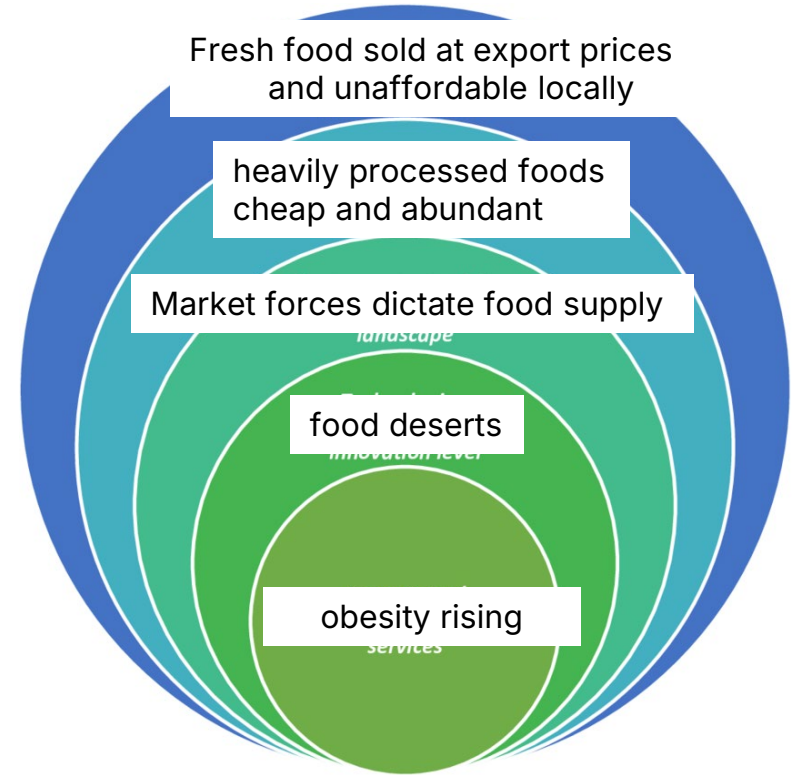
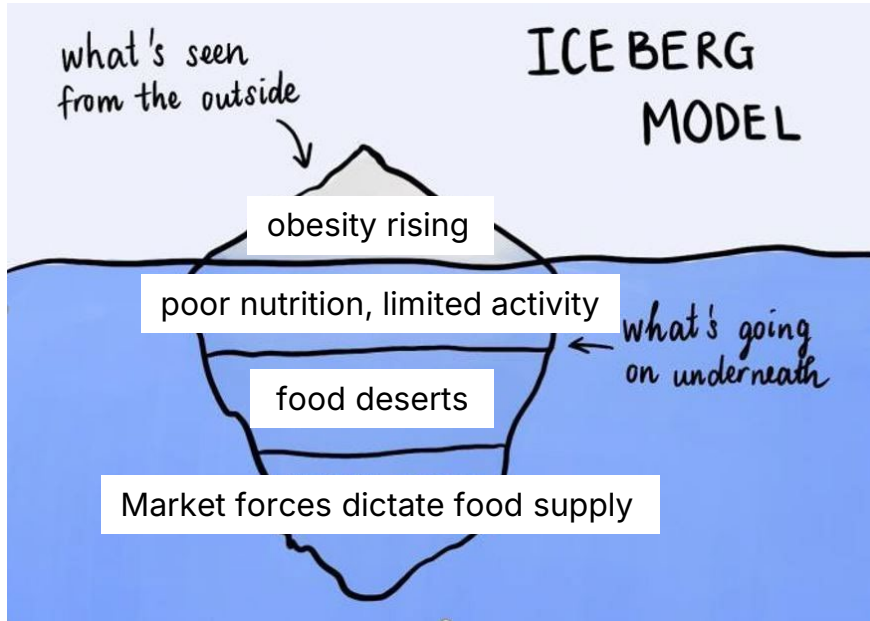
Obesity



Obesity



Obesity





Tools for ST Exploration

Iceberg Model

Levels of Thinking

Tools for ST Exploration

Iceberg Model

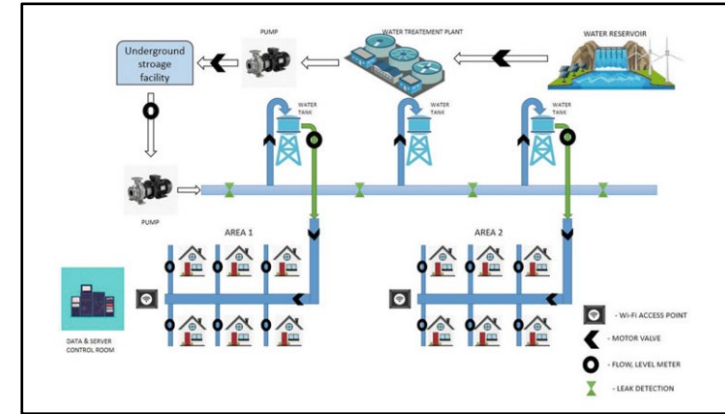
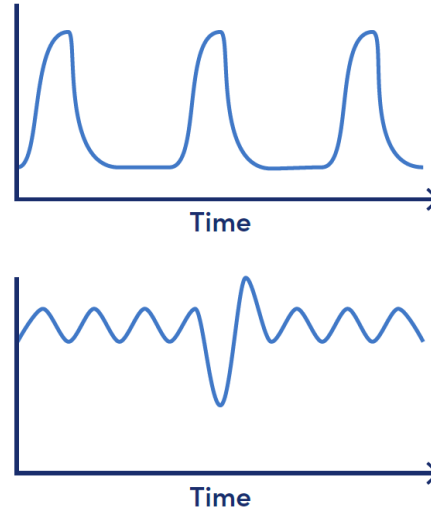
Levels of Thinking

Trend Maps

System Models

Conceptual

Computational





Questions into the ST tools

What happened?

What recurring events are we noticing?

What length of time is long enough to see patterns in systems function?

What structures may be determining the function we see?

What underlying beliefs, values, or assumptions are at play?

What are the feedbacks in this system?

How can we define a boundary around this system?

What other perspectives, methodologies, or disciplines might help us more fully understand the issue?

Where in this system could we make small shifts that would make big difference? What trade-offs exist? Is there a potential for unintended consequences?



Tools for ST Exploration

Iceberg Model

Levels of Thinking

Trend Maps

System Models

Conceptual

Computational

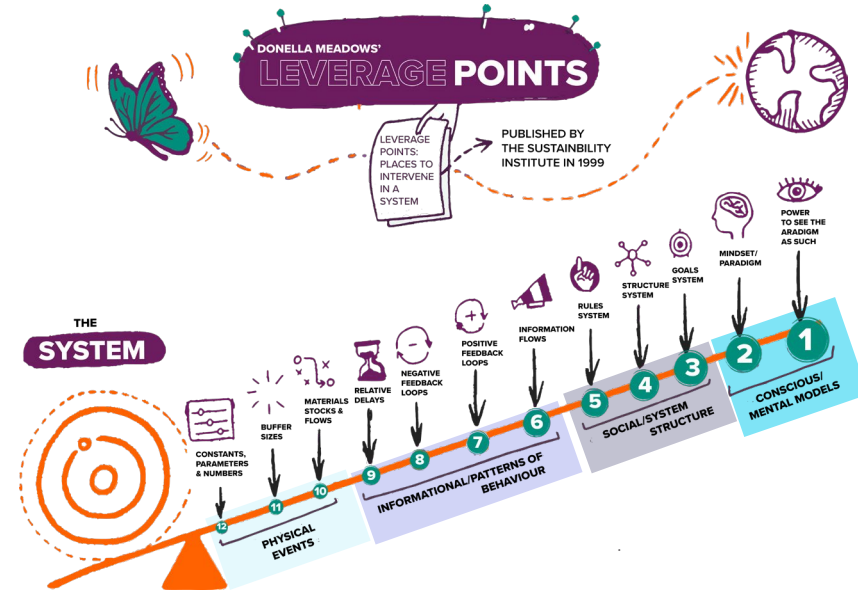
Leverage Points

Causal Loops

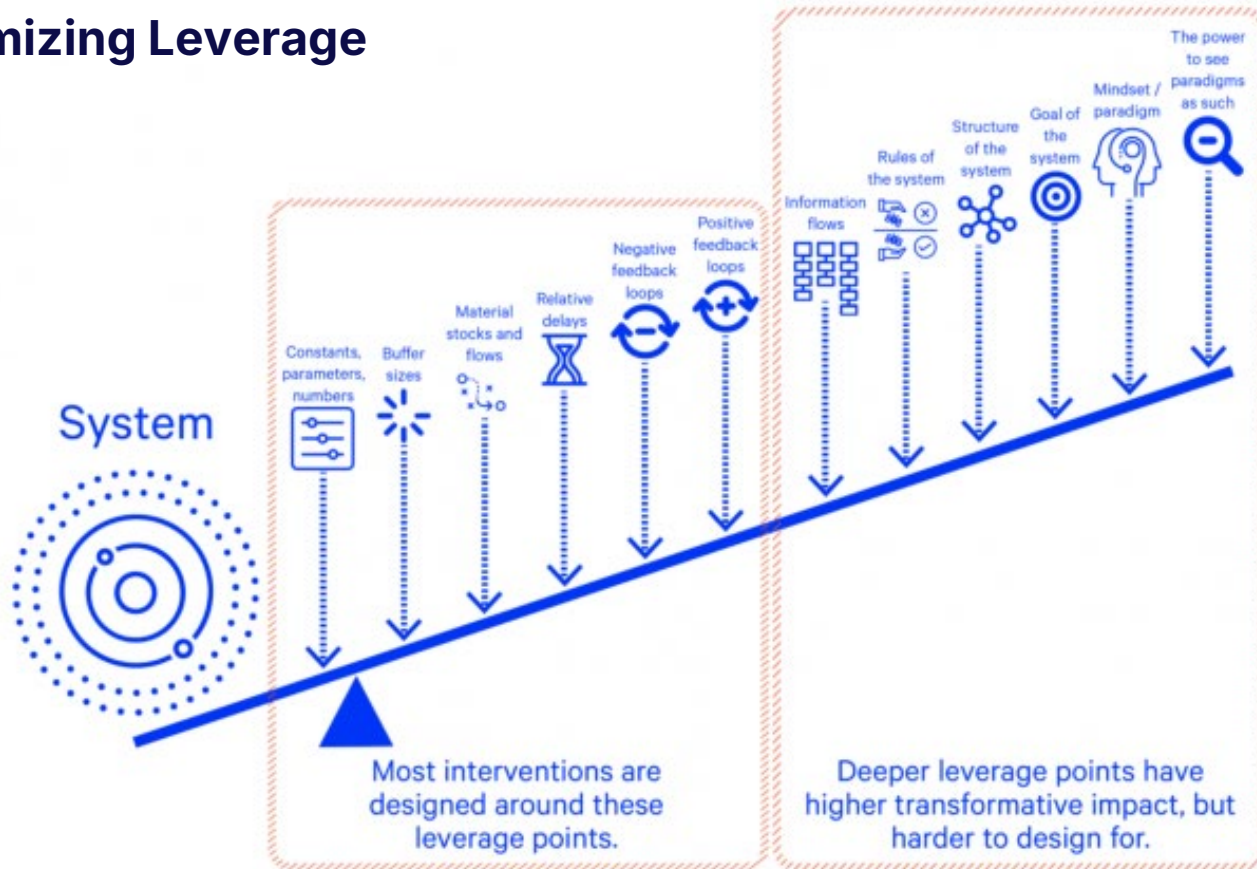
Feedback Loops

Leverage Points

12. Constants, parameters, numbers (such as subsidies, taxes, standards).
11. The sizes of buffers and other stabilizing stocks, relative to their flows.
10. The structure of material stocks and flows (such as transport networks, population age structures).
9. The lengths of delays, relative to the rate of system change.
8. The strength of negative feedback loops, relative to the impacts they are trying to correct against.
7. The gain around driving positive feedback loops.
6. The structure of information flows (who does and does not have access to information).
5. The rules of the system (such as incentives, punishments, constraints).
4. The power to add, change, evolve, or self-organize system structure.
3. The goals of the system.
2. The mindset or paradigm out of which the system — its goals, structure, rules, delays, parameters — arises.
1. The power to transcend paradigms.



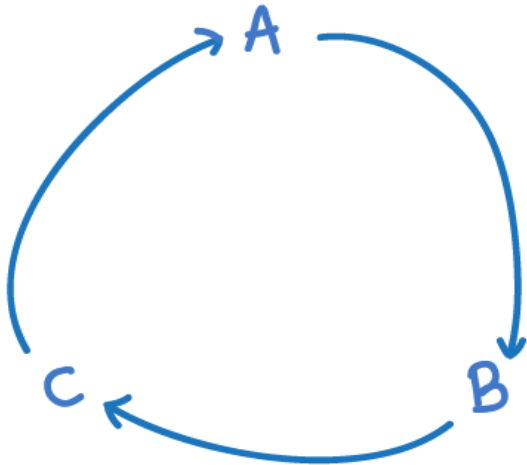
Maximizing Leverage



Adapted from Meadows (1977): Leverage Points: Places to Intervene in a System

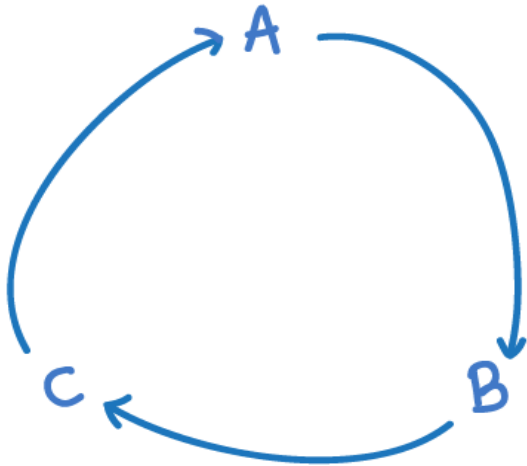
Causal Loops & Feedback Loops

Feedback Loop

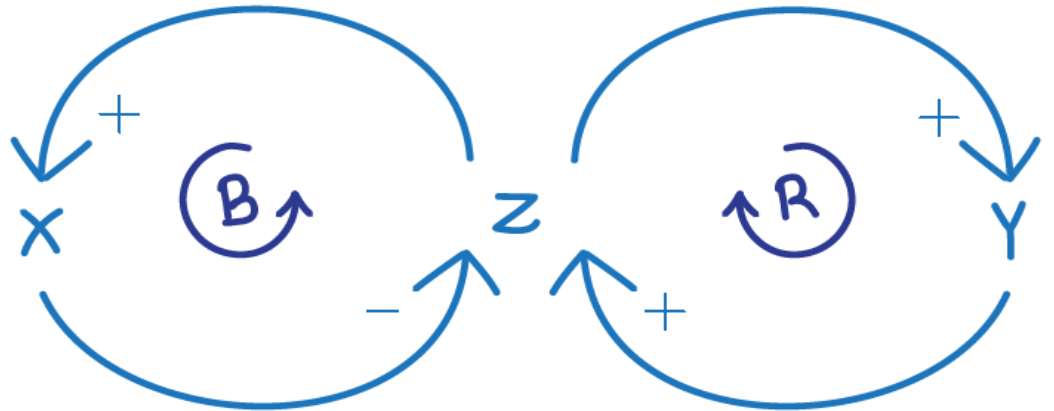


Causal Loops & Feedback Loops

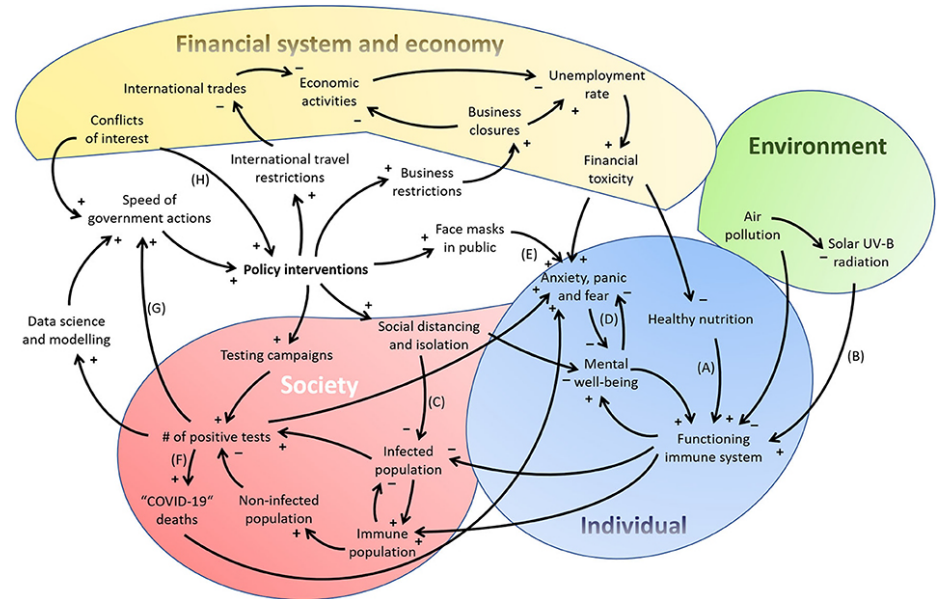
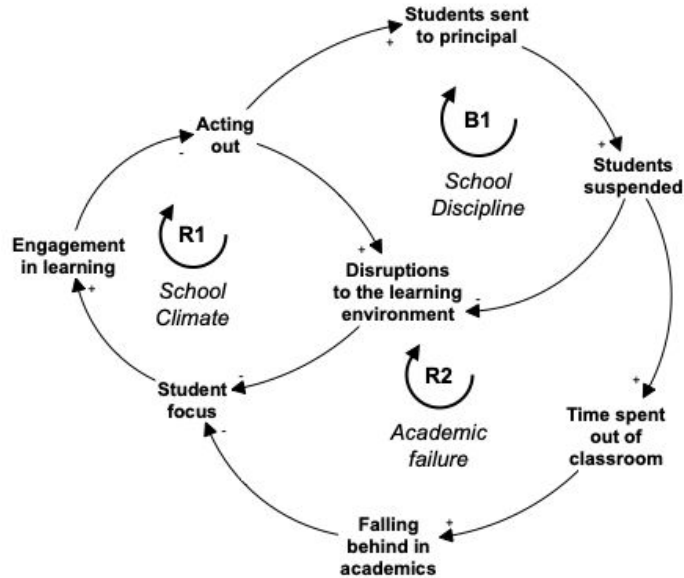
Feedback Loop



Causal Loop Diagram



Causal Loops





Why Systems Work

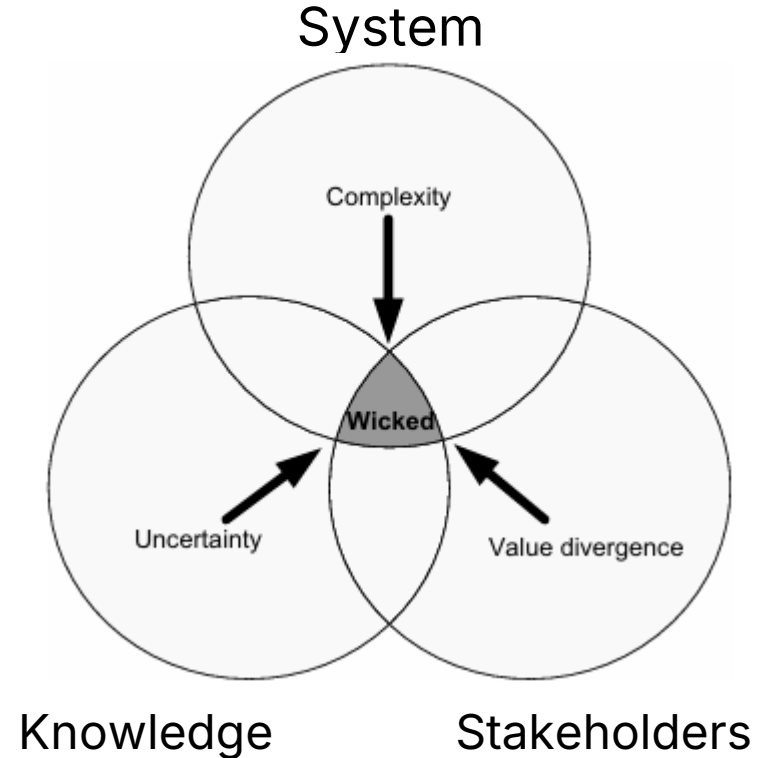
Robust systems tend to work pretty well (and can be hard to disrupt!)

- Resilience
- Self Organise
- Hierarchy

To intervene in the system, identify Leverage Points or Causal Loops to impact

Summary

- System: elements, relationships, purpose
- Complexity and "Wickedness"
- Systems Thinking is NOT an Algorithm or Single Process!
- Its approach to broader understanding, critical thinking, and proposing of interventions and solutions.





Questions?

Thank you, and see you next
lecture 😊

